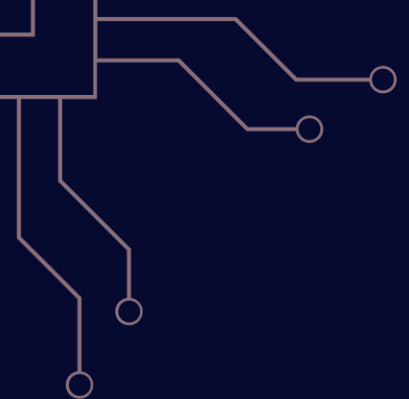
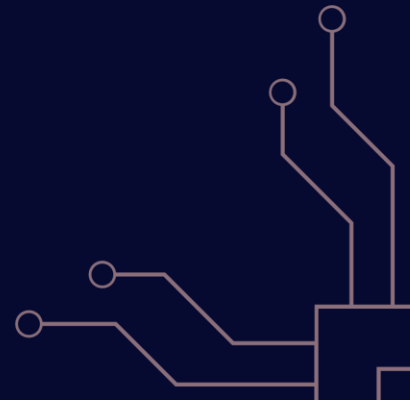




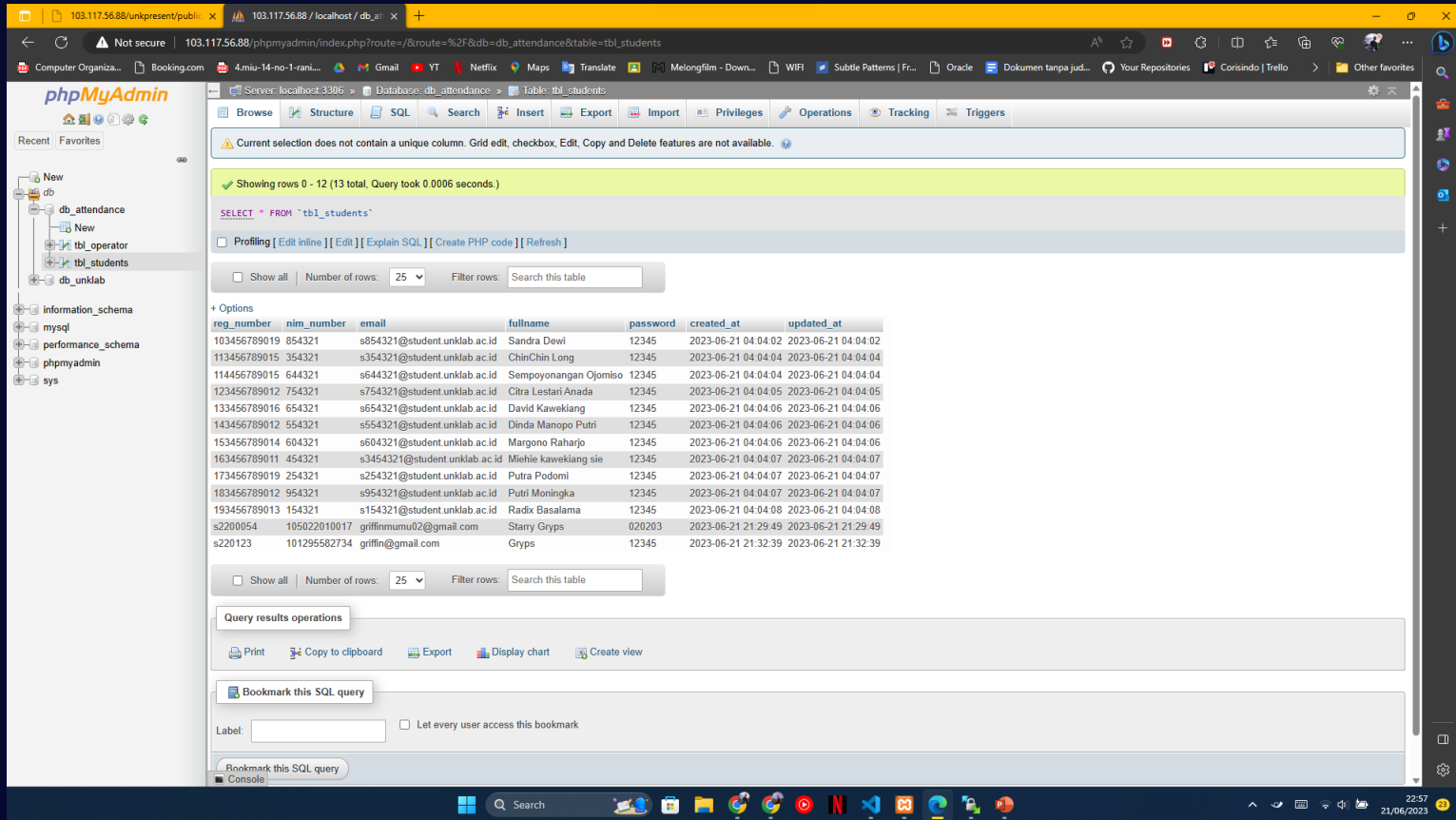
GRIFFIN M. Mumu



Exercise #1

DevOps

Ubah Data dalam Database

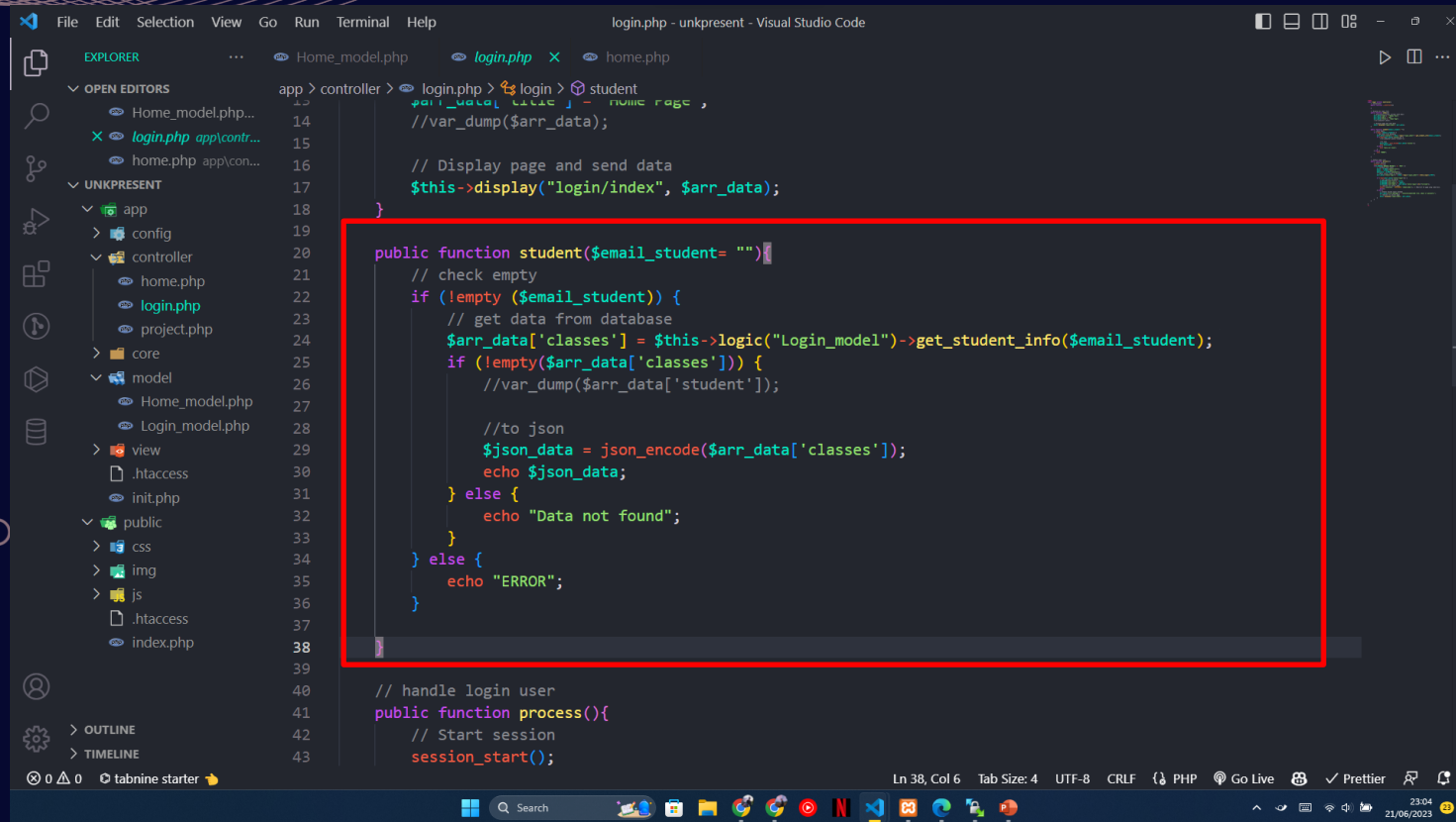


The screenshot shows the phpMyAdmin web interface. The left sidebar displays the database structure with 'db' selected, containing 'tbl_students'. The main panel shows the 'tbl_students' table with 13 rows of data. The table has columns: reg_number, nim_number, email, fullname, password, created_at, and updated_at. The data includes 11 existing rows and 2 newly added rows at the bottom.

reg_number	nim_number	email	fullname	password	created_at	updated_at
103456789019	854321	s854321@student.unklab.ac.id	Sandra Dewi	12345	2023-06-21 04:04:02	2023-06-21 04:04:02
113456789015	354321	s354321@student.unklab.ac.id	ChinChin Long	12345	2023-06-21 04:04:04	2023-06-21 04:04:04
114456789015	644321	s644321@student.unklab.ac.id	Sempoyonangan Ojomiso	12345	2023-06-21 04:04:04	2023-06-21 04:04:04
123456789012	754321	s754321@student.unklab.ac.id	Citra Lestari Anada	12345	2023-06-21 04:04:05	2023-06-21 04:04:05
133456789016	654321	s654321@student.unklab.ac.id	David Kawekiang	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
143456789012	554321	s554321@student.unklab.ac.id	Dinda Manopo Putri	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
153456789014	604321	s604321@student.unklab.ac.id	Margono Raharjo	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
163456789011	454321	s3454321@student.unklab.ac.id	Miehie kawekiang sie	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
173456789019	254321	s254321@student.unklab.ac.id	Putra Podomi	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
183456789012	954321	s954321@student.unklab.ac.id	Putri Moningka	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
193456789013	154321	s154321@student.unklab.ac.id	Radix Basalama	12345	2023-06-21 04:04:08	2023-06-21 04:04:08
s2200054	105022010017	griffinumu02@gmail.com	Starry Gryps	020203	2023-06-21 21:29:49	2023-06-21 21:29:49
s220123	101295582734	griffin@gmail.com	Gryps	12345	2023-06-21 21:32:39	2023-06-21 21:32:39

Disini kita mengubah data yang ada pada table "tbl_students" yang awalnya hanya 5 data, sekarang ada 11 data baru (2 data terakhir adalah data hasil input)

Tambahkan function di login.php



```
login.php - unkpresent - Visual Studio Code

EXPLORER
  app > controller > login.php
  Home_model.php
  login.php
  home.php

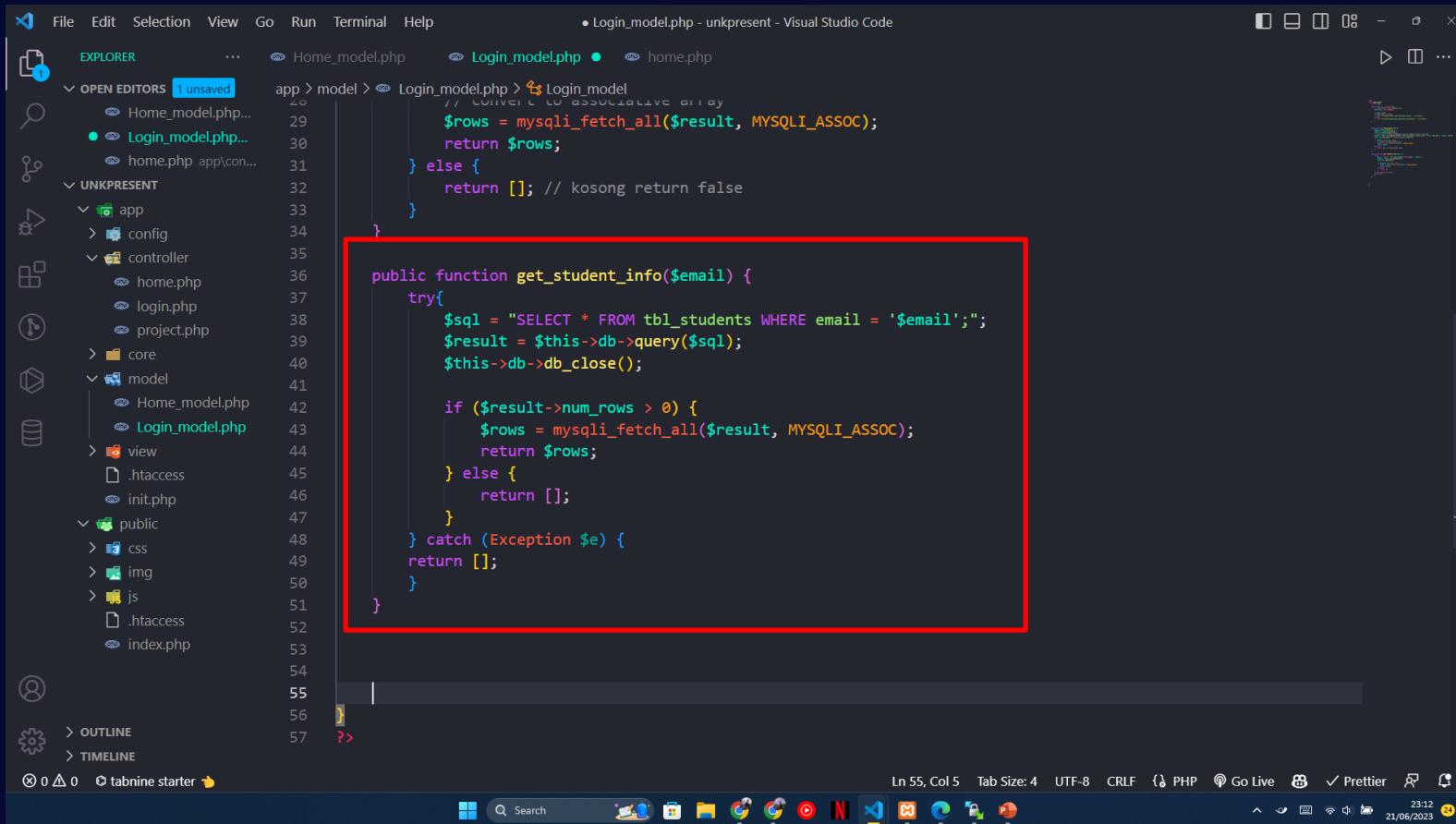
OPEN EDITORS
  app > controller > login.php
  Home_model.php
  login.php
  home.php

UNKPRESENT
  app
  config
  controller
  home.php
  login.php
  project.php
  core
  model
  Home_model.php
  Login_model.php
  view
  .htaccess
  init.php
  public
  css
  img
  js
  .htaccess
  index.php

app > controller > login.php
14 $arr_data['title'] = 'home page';
15 //var_dump($arr_data);
16
17 // Display page and send data
18 $this->display("login/index", $arr_data);
19
20
21
22
23 public function student($email_student = ""){
24     // check empty
25     if (empty($email_student)) {
26         // get data from database
27         $arr_data['classes'] = $this->logic("Login_model")->get_student_info($email_student);
28         if (empty($arr_data['classes'])) {
29             //var_dump($arr_data['student']);
30
31             //to json
32             $json_data = json_encode($arr_data['classes']);
33             echo $json_data;
34         } else {
35             echo "Data not found";
36         }
37     } else {
38         echo "ERROR";
39     }
40 }
41
42 // handle login user
43 public function process(){
44     // Start session
45     session_start();
46 }
```

Fungsi dari public function student dalam kode tersebut adalah untuk menangani permintaan terkait data mahasiswa. Fungsi ini menerima parameter opsional yaitu \$email_student yang digunakan untuk mengidentifikasi mahasiswa yang ingin ditampilkan datanya. Function ini yang akan memanggil model function untuk mengambil data di database berdasarkan email dari mahasiswa.

Tambahkan function di 'Login_model.php'



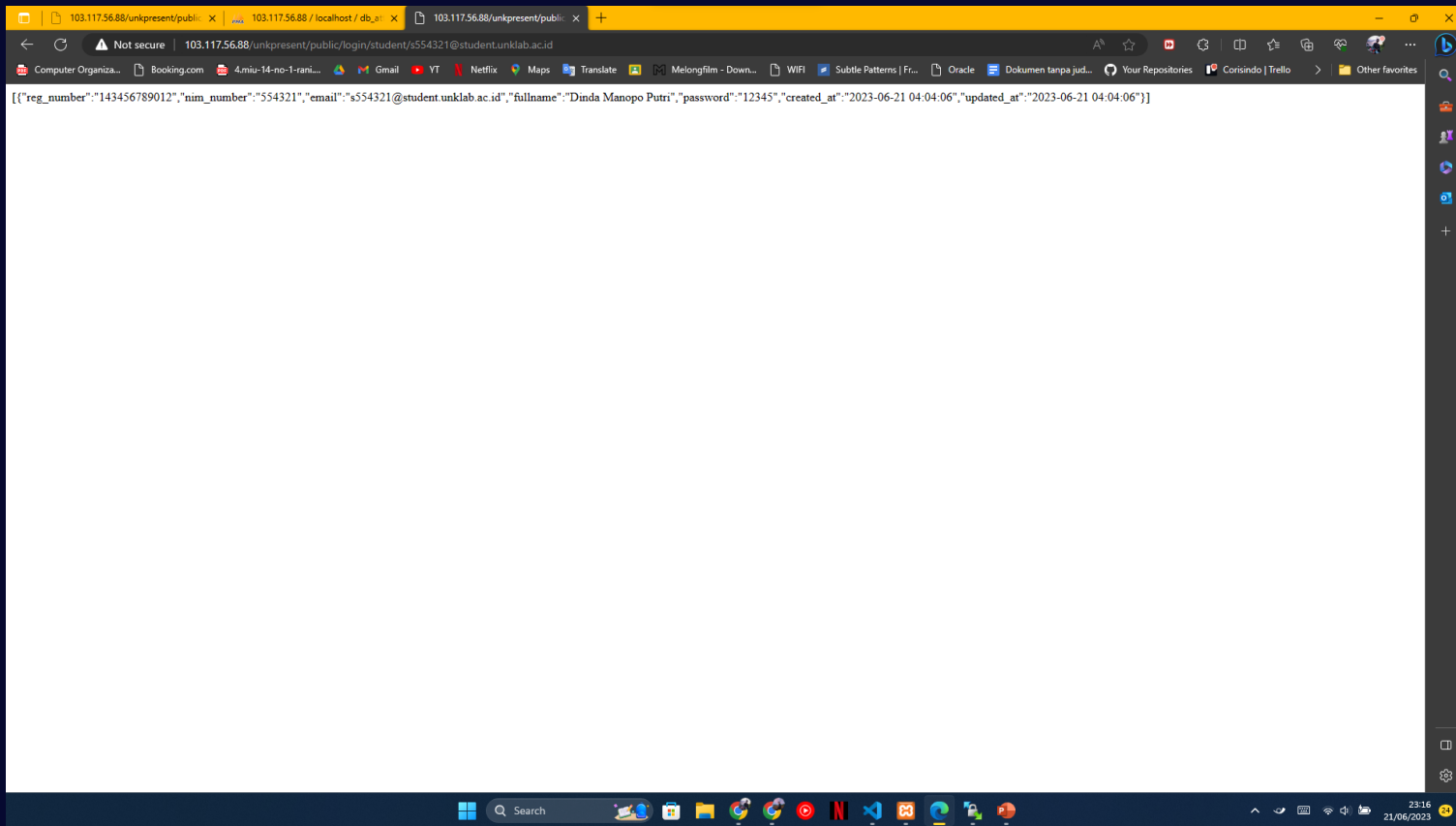
The screenshot shows the Visual Studio Code editor with the file 'Login_model.php' open. The Explorer sidebar on the left shows the project structure, including folders for 'app', 'controller', 'core', 'model', 'view', and 'public'. The 'model' folder is expanded, showing 'Home_model.php' and 'Login_model.php'. The main editor area displays the code for 'Login_model.php'. A red rectangle highlights a new function being added to the file. The function is named 'get_student_info' and takes an '\$email' parameter. It uses a try-catch block to execute a SQL query that selects all data from the 'tbl_students' table where the email matches the provided '\$email'. If the query is successful and returns rows, it fetches all results and returns them as an associative array. If there is an error or no rows are found, it returns an empty array. The status bar at the bottom indicates the cursor is at line 55, column 5.

```
Ln 55, Col 5  Tab Size: 4  UTF-8  CRLF  { PHP  Go Live  Prettier
```

```
public function get_student_info($email) {  
    try{  
        $sql = "SELECT * FROM tbl_students WHERE email = '$email'";  
        $result = $this->db->query($sql);  
        $this->db->db_close();  
  
        if ($result->num_rows > 0) {  
            $rows = mysqli_fetch_all($result, MYSQLI_ASSOC);  
            return $rows;  
        } else {  
            return [];  
        }  
    } catch (Exception $e) {  
        return [];  
    }  
}
```

Kemudian kita menambahkan metode baru bernama "get_student_info" di dalam model "Login" pada file "Model_login.php". Metode ini akan digunakan untuk mengambil data secara langsung dari database berdasarkan email mahasiswa.

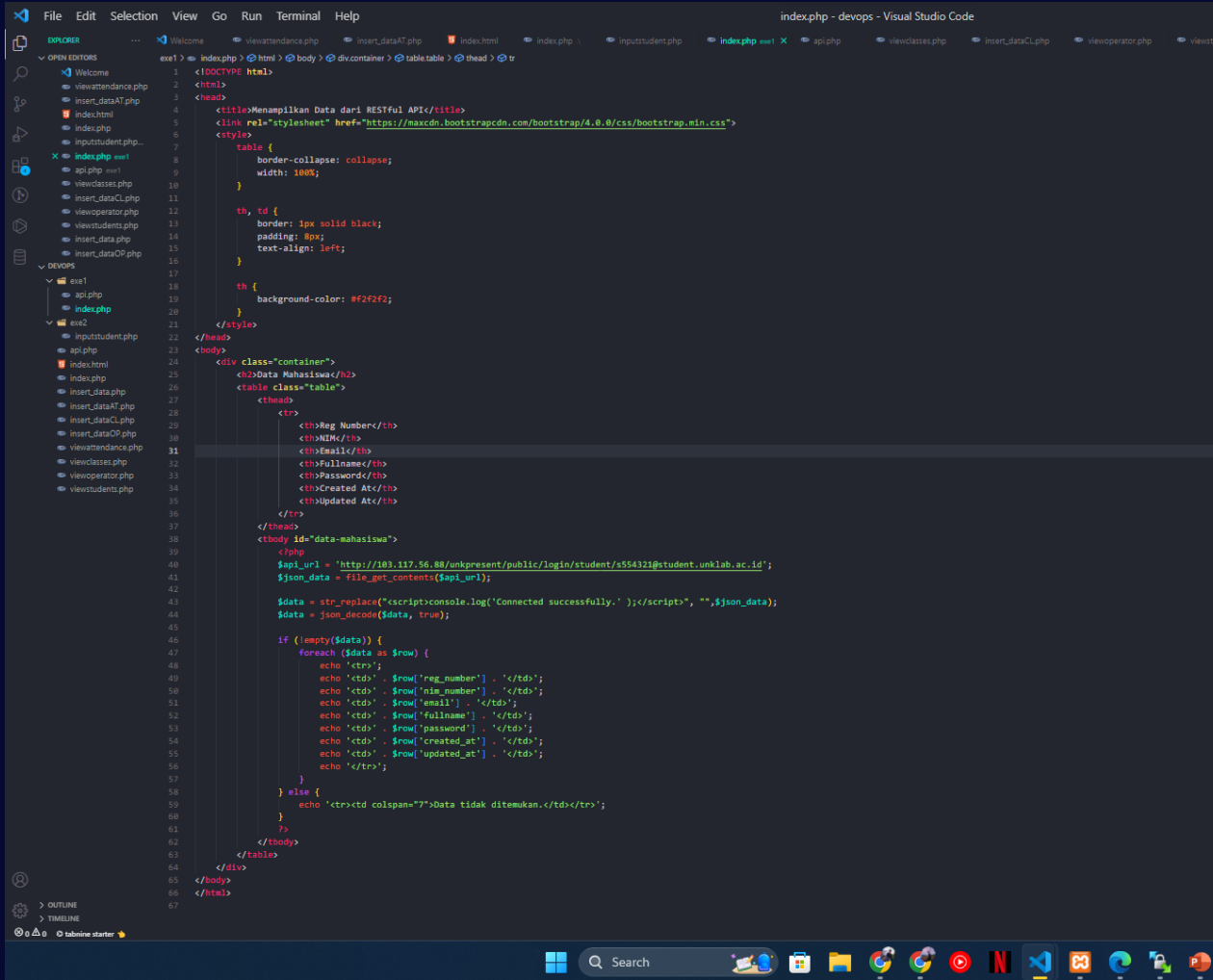
Output from RESTful API



Output dari RESTful API tersebut berbentuk JSON format dan akan menampilkan data dari mahasiswa sesuai dengan email yang dimasukan

Display Output in Table HTML

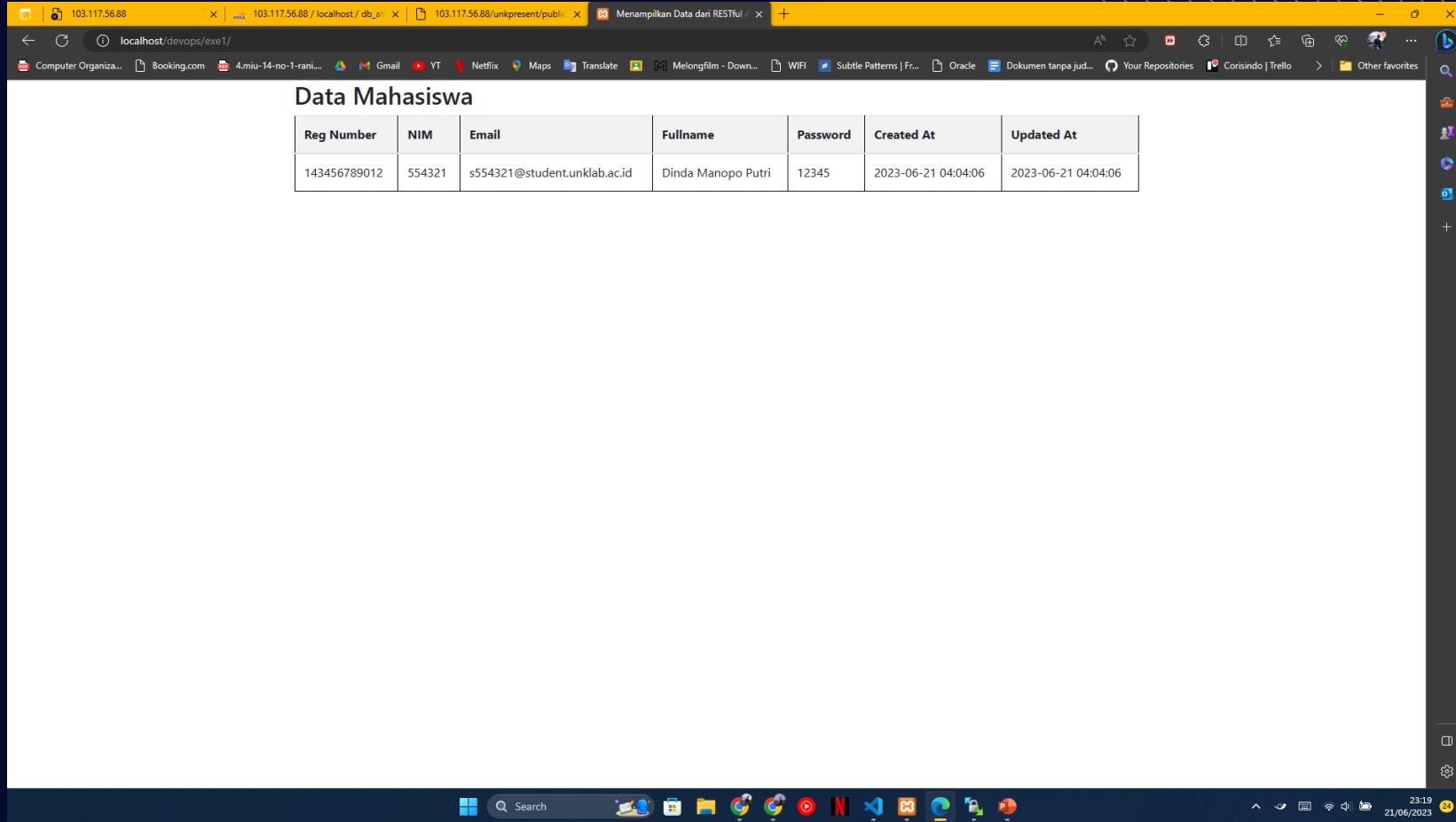
Buat Code di local PC untuk menangkap dan menampilkan data dari RESTful API tersebut di dalam table HTML.



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Menampilkan Data dari RESTful API</title>
5 <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.0.0/css/bootstrap.min.css">
6 </head>
7 <body>
8 <div class="container">
9 <div class="table">
10 <table>
11 <thead>
12 <tr>
13 <th>Reg Number</th>
14 <th>NIM</th>
15 <th>Email</th>
16 <th>Fullname</th>
17 <th>Password</th>
18 <th>Created At</th>
19 <th>Updated At</th>
20 </tr>
21 </thead>
22 <tbody>
23 <tr>
24 <td>Data Mahasiswa</td>
25 </tr>
26 </tbody>
27 </table>
28 </div>
29 </div>
30 </body>
31 </html>
```

The code is a PHP script that fetches data from a RESTful API and displays it in an HTML table. The table has columns for Reg Number, NIM, Email, Fullname, Password, Created At, and Updated At. The data is fetched from the API using the `file\_get\_contents` function and decoded using `json\_decode`. The data is then iterated over using a `foreach` loop to generate the table rows. If no data is found, a message is displayed.

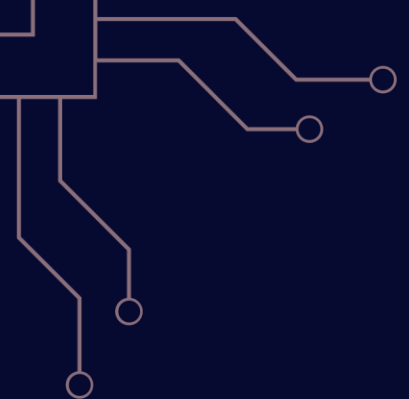
Display Output in Table HTML



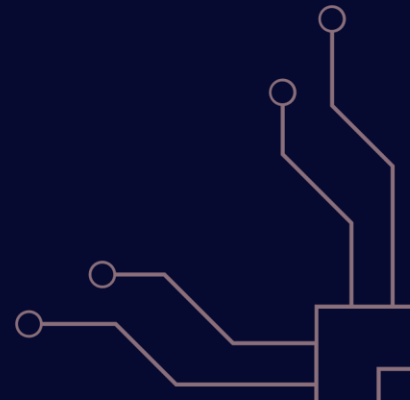
The screenshot shows a web browser window with a yellow address bar and a dark sidebar on the right. The main content area displays a table titled "Data Mahasiswa". The table has seven columns: Reg Number, NIM, Email, Fullname, Password, Created At, and Updated At. A single row of data is visible, representing a student named Dinda Manopo Putri.

Reg Number	NIM	Email	Fullname	Password	Created At	Updated At
143456789012	554321	s554321@student.unklab.ac.id	Dinda Manopo Putri	12345	2023-06-21 04:04:06	2023-06-21 04:04:06

Ini adalah output dari code yang telah dibuat tadi,
yang mana menampilkan data JSON nya ke dalam
table html



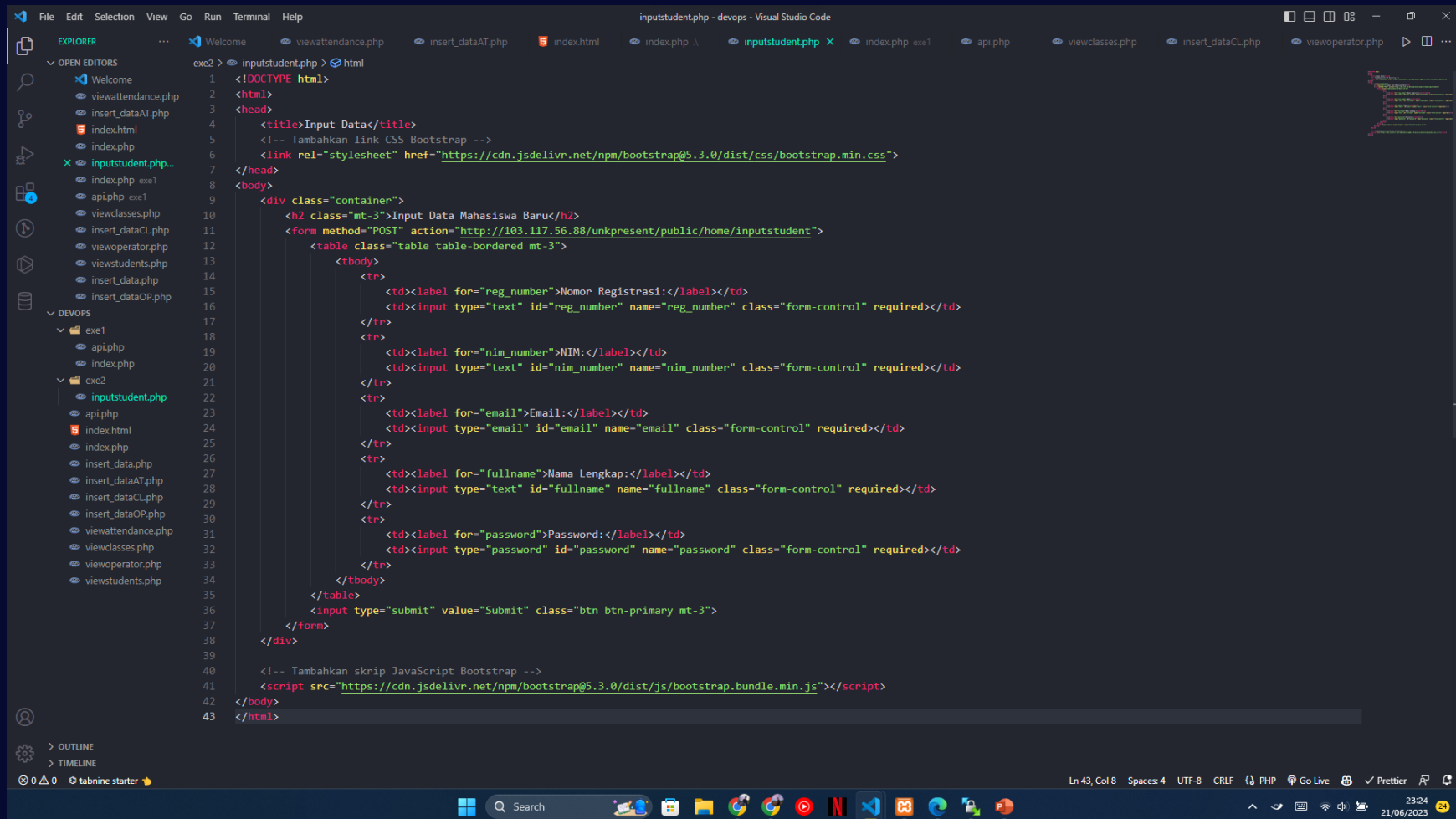
GRIFFIN M. Mumu



Exercise #2

DevOps

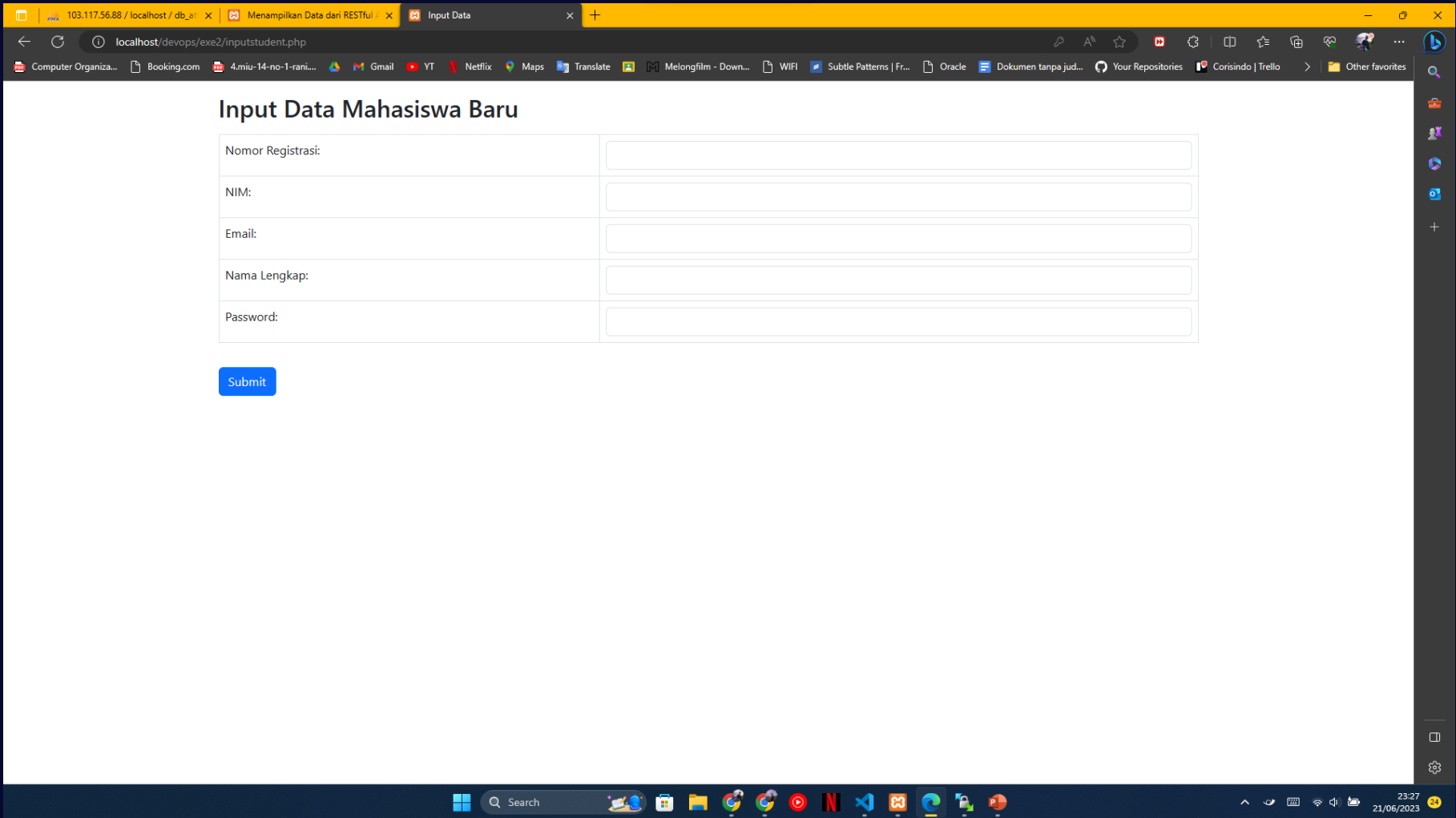
Buat Form input data di local PC



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Input Data</title>
5 <!-- Tambahkan link CSS Bootstrap -->
6 <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
7 </head>
8 <body>
9 <div class="container">
10 <h2 class="mt-3">Input Data Mahasiswa Baru</h2>
11 <form method="POST" action="http://103.117.56.88/unkpresent/public/home/inputstudent">
12 <table class="table table-bordered mt-3">
13 <tbody>
14 <tr>
15 <td><label for="reg_number">Nomor Registrasi:</label></td>
16 <td><input type="text" id="reg_number" name="reg_number" class="form-control" required></td>
17 </tr>
18 <tr>
19 <td><label for="nim_number">NIM:</label></td>
20 <td><input type="text" id="nim_number" name="nim_number" class="form-control" required></td>
21 </tr>
22 <tr>
23 <td><label for="email">Email:</label></td>
24 <td><input type="email" id="email" name="email" class="form-control" required></td>
25 </tr>
26 <tr>
27 <td><label for="fullname">Nama Lengkap:</label></td>
28 <td><input type="text" id="fullname" name="fullname" class="form-control" required></td>
29 </tr>
30 <tr>
31 <td><label for="password">Password:</label></td>
32 <td><input type="password" id="password" name="password" class="form-control" required></td>
33 </tr>
34 </tbody>
35 </table>
36 <input type="submit" value="Submit" class="btn btn-primary mt-3">
37 </form>
38 </div>
39
40 <!-- Tambahkan skrip JavaScript Bootstrap -->
41 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
42 </body>
43 </html>
```

Form tersebut kemudian akan digunakan untuk mengirim data lewat metode “post” melalui API

Preview in Browser



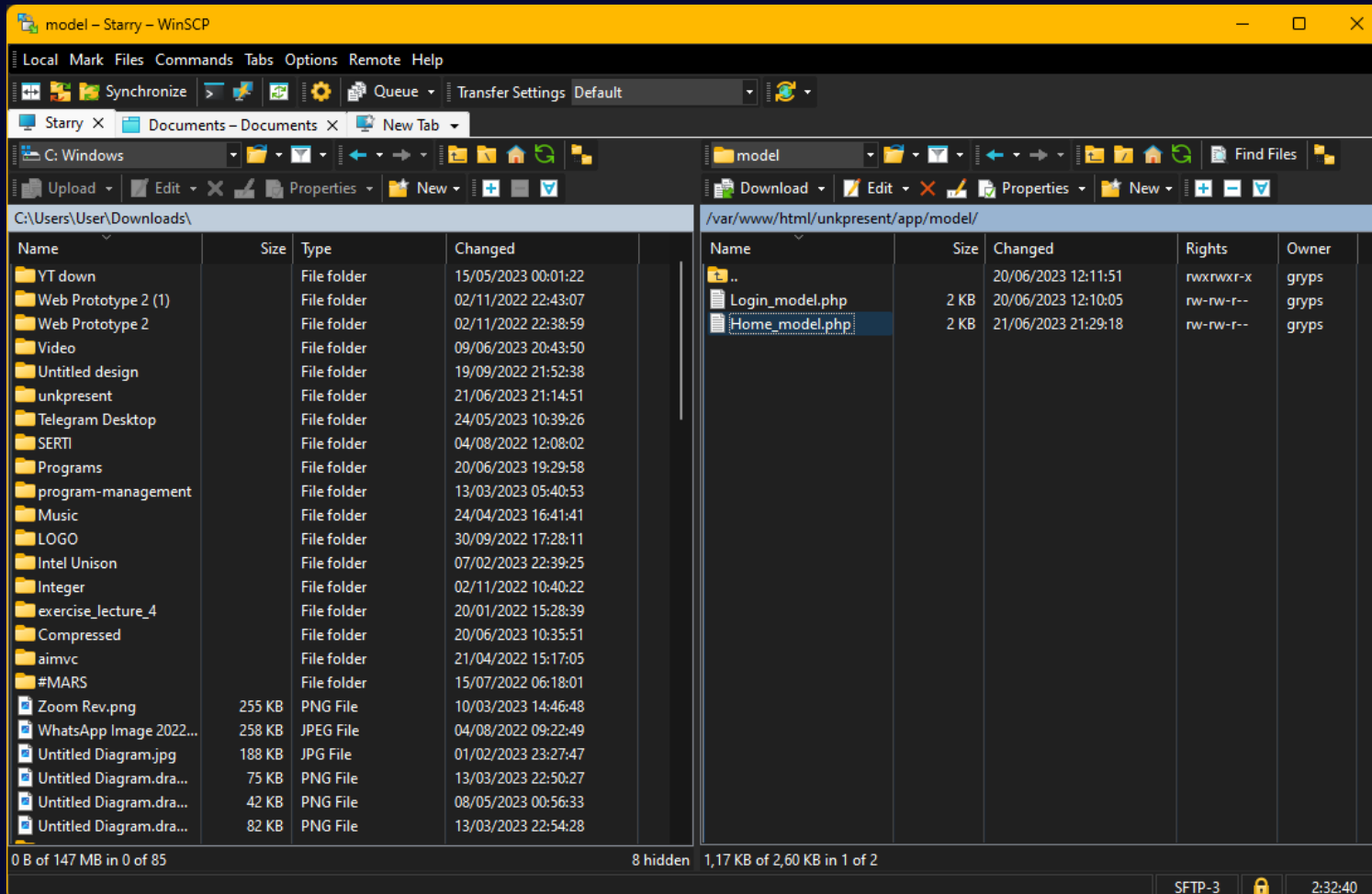
The screenshot shows a web browser window with the address bar displaying `localhost/devops/exe2/inputstudent.php`. The page title is "Input Data Mahasiswa Baru". The form contains five input fields and a submit button:

Nomor Registrasi:	
NIM:	
Email:	
Nama Lengkap:	
Password:	

Below the form is a blue "Submit" button.

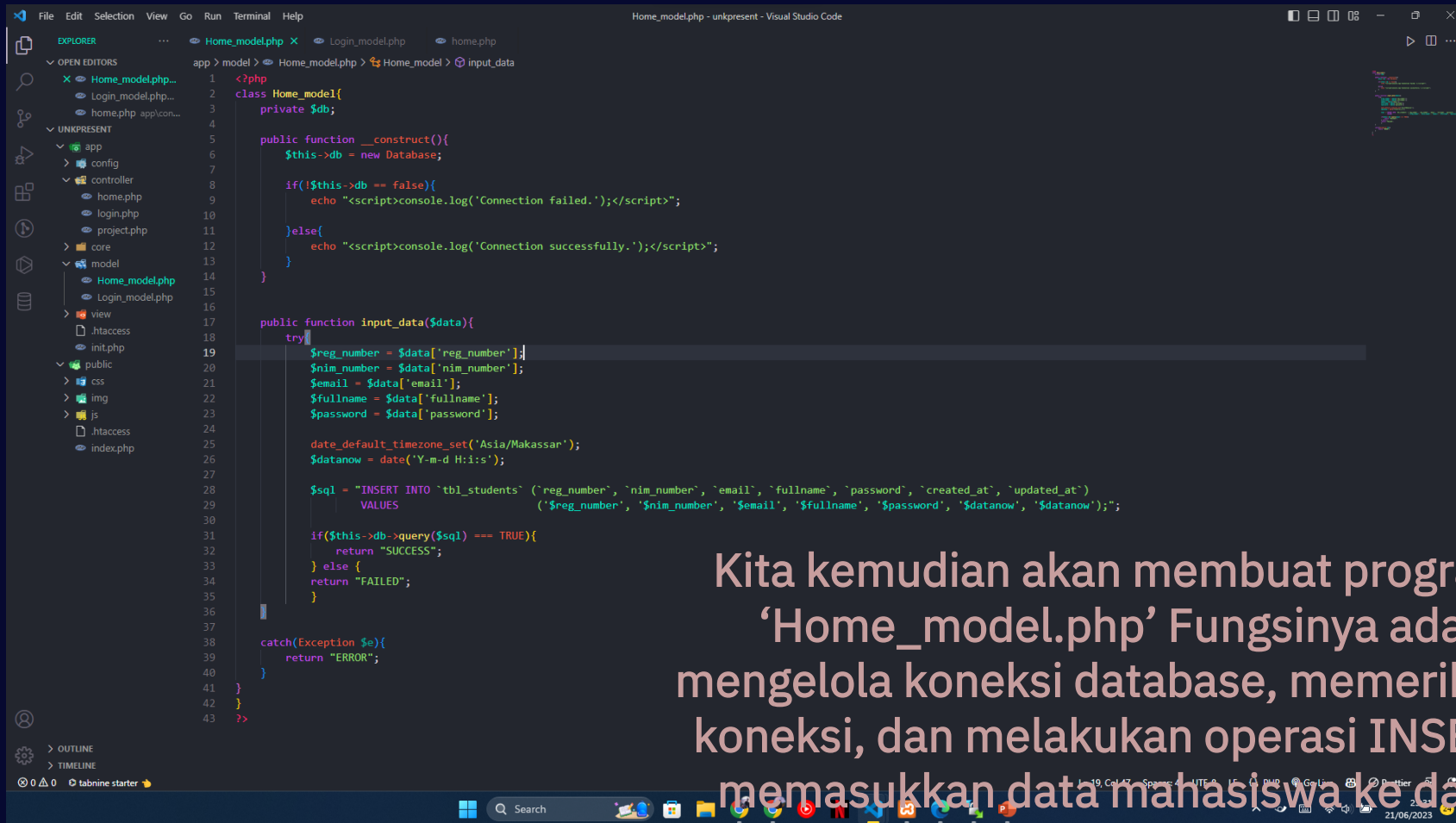
Ini adalah preview dari form yang sudah kita coding di atas yang mana tersimpan di local pc kita

Create New Model File



Kita kemudian membuat sebuah model file di directory “model” dengan nama “Home_model.php”

Add Script to "Home_model.php"



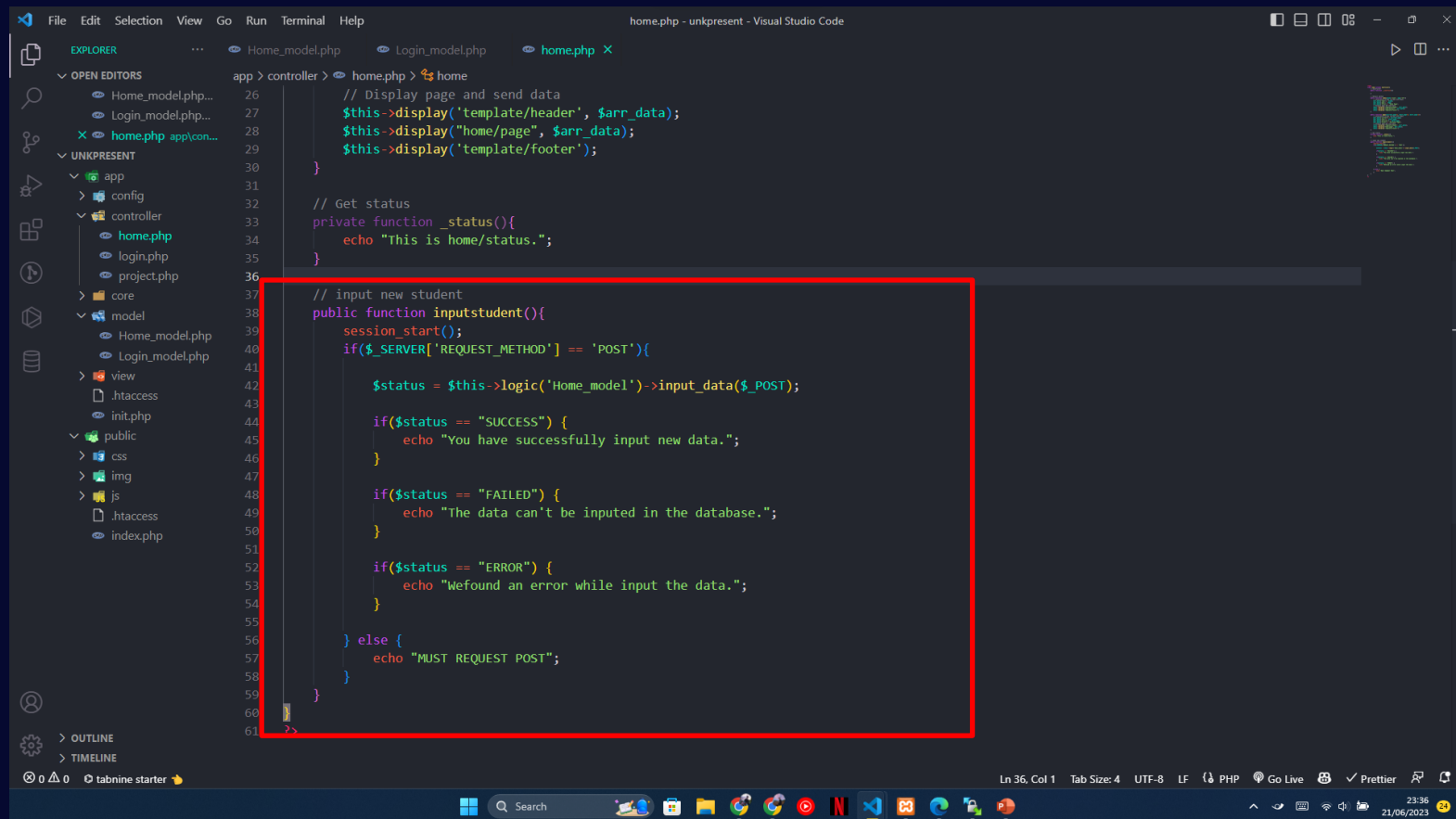
```
<?php
class Home_model{
    private $db;

    public function __construct(){
        $this->db = new Database;

        if(!$this->db == false){
            echo "<script>console.log('Connection failed.');
```

Kita kemudian akan membuat program dalam 'Home_model.php' Fungsinya adalah untuk mengelola koneksi database, memeriksa status koneksi, dan melakukan operasi INSERT untuk memasukkan data mahasiswa ke dalam tabel "tbl_students". Metode konstruktor digunakan untuk membuat objek koneksi database, dan di dalamnya dilakukan pengecekan status koneksi. Metode "input_data" digunakan untuk menginputkan data mahasiswa ke dalam tabel dengan menggunakan pernyataan SQL INSERT.

Add Function to Controller "Home.php"



```
26 // Display page and send data
27 $this->display('template/header', $arr_data);
28 $this->display("home/page", $arr_data);
29 $this->display('template/footer');
30 }
31
32 // Get status
33 private function _status(){
34     echo "This is home/status.";
35 }
36
37 // input new student
38 public function inputstudent(){
39     session_start();
40     if($_SERVER['REQUEST_METHOD'] == 'POST'){
41
42         $status = $this->logic('Home_model')->input_data($_POST);
43
44         if($status == "SUCCESS") {
45             echo "You have successfully input new data.";
46         }
47
48         if($status == "FAILED") {
49             echo "The data can't be inputed in the database.";
50         }
51
52         if($status == "ERROR") {
53             echo "We found an error while input the data.";
54         }
55
56     } else {
57         echo "MUST REQUEST POST";
58     }
59 }
60
61 }
```

fungsi `inputstudent()` digunakan untuk mengelola permintaan input data mahasiswa baru melalui metode POST. Ia memanggil metode model `input_data()` untuk memproses data dan memberikan respons sesuai dengan hasil operasi yang dilakukan terhadap database.

Input New Student Data

Input Data Mahasiswa Baru

Nomor Registrasi:	s12345
NIM:	7777777777
Email:	stary14@gmail.com
Nama Lengkap:	Griffin Mumu
Password:	*****

Submit

You have successfully input new data.

phpMyAdmin

Showing rows 0 - 14 (15 total, Query took 0.0006 seconds)

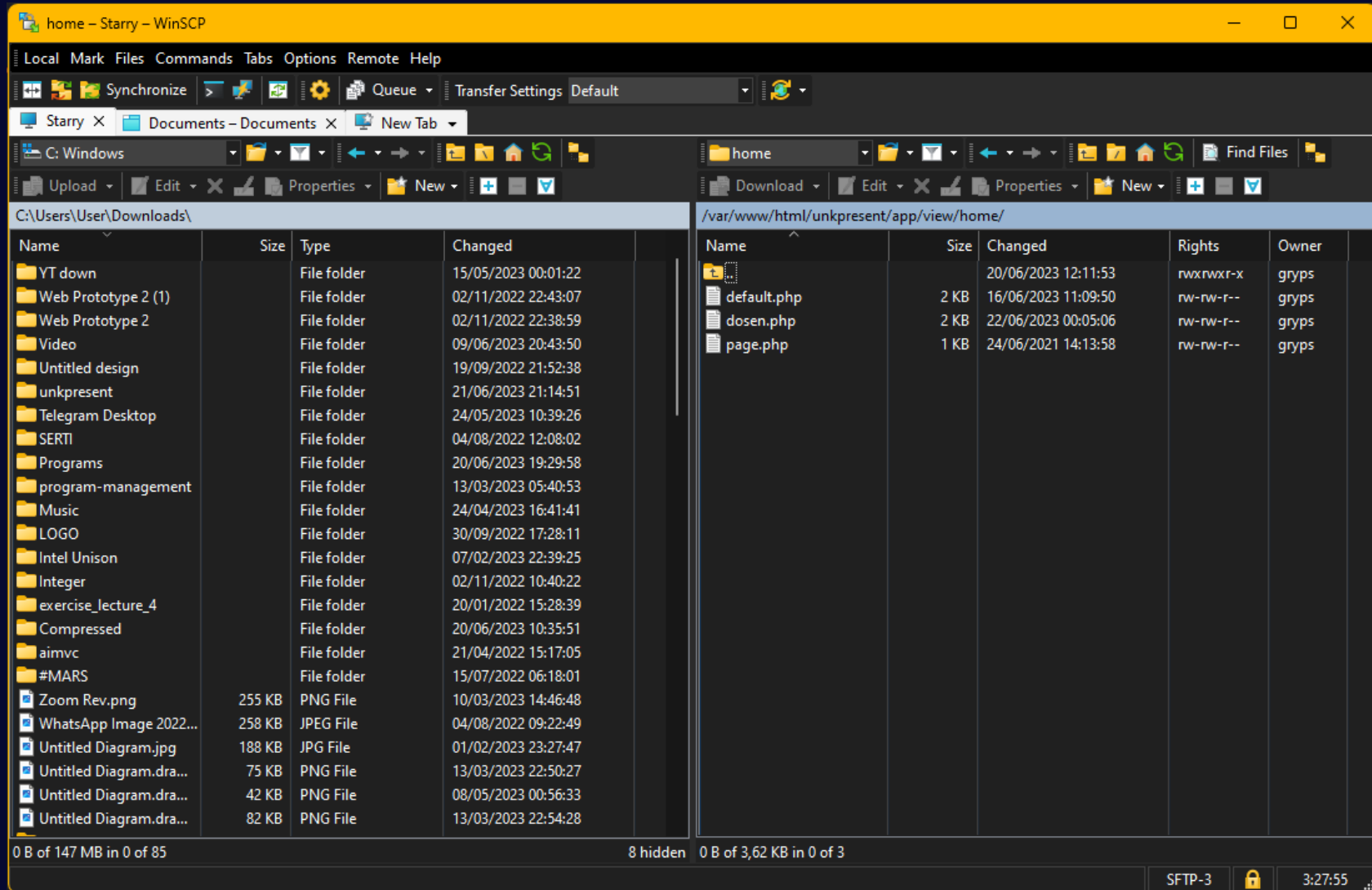
SELECT * FROM `tbl\_students`

reg_number	nim_number	email	fullname	password	created_at	updated_at
103456789019	854321	a854321@student.unklab.ac.id	Sandra Dewi	12345	2023-06-21 04:04:02	2023-06-21 04:04:02
113456789015	354321	s354321@student.unklab.ac.id	ChinChin Long	12345	2023-06-21 04:04:04	2023-06-21 04:04:04
114456789015	644321	a644321@student.unklab.ac.id	Sempoyonangan Ojoniwo	12345	2023-06-21 04:04:04	2023-06-21 04:04:04
123456789012	754321	a754321@student.unklab.ac.id	Citra Lestari Ananda	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
133456789016	654321	a654321@student.unklab.ac.id	David Kawekiang	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
143456789012	554321	a554321@student.unklab.ac.id	Dinda Manopo Putri	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
153456789014	604321	a604321@student.unklab.ac.id	Margono Raharjo	12345	2023-06-21 04:04:06	2023-06-21 04:04:06
163456789011	454321	s3454321@student.unklab.ac.id	Mohie kawekiang sie	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
173456789019	254321	a254321@student.unklab.ac.id	Putra Podomi	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
183456789012	954321	a954321@student.unklab.ac.id	Putri Moringka	12345	2023-06-21 04:04:07	2023-06-21 04:04:07
193456789013	154321	s154321@student.unklab.ac.id	Radix Basalama	12345	2023-06-21 04:04:08	2023-06-21 04:04:08
s2200054	105022010017	griffinmumu02@gmail.com	Stary Gryps	026203	2023-06-21 21:29:49	2023-06-21 21:29:49
s220123	101295682734	griffin@gmail.com	Gryps	12345	2023-06-21 21:32:39	2023-06-21 21:32:39
s12345	7777777777	stary14@gmail.com	Griffin Mumu	026203	2023-06-21 23:41:01	2023-06-21 23:41:01

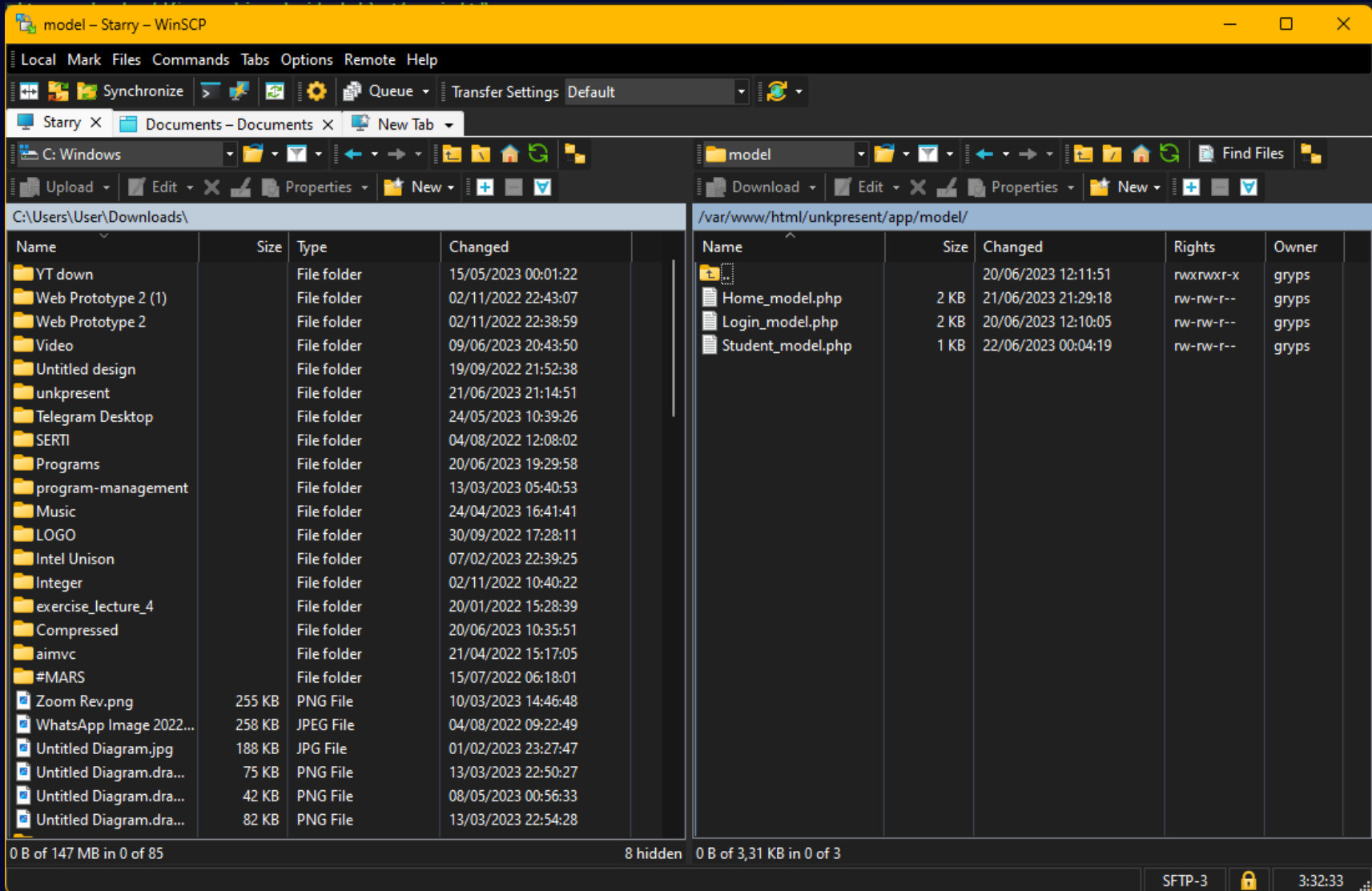
Thank You.



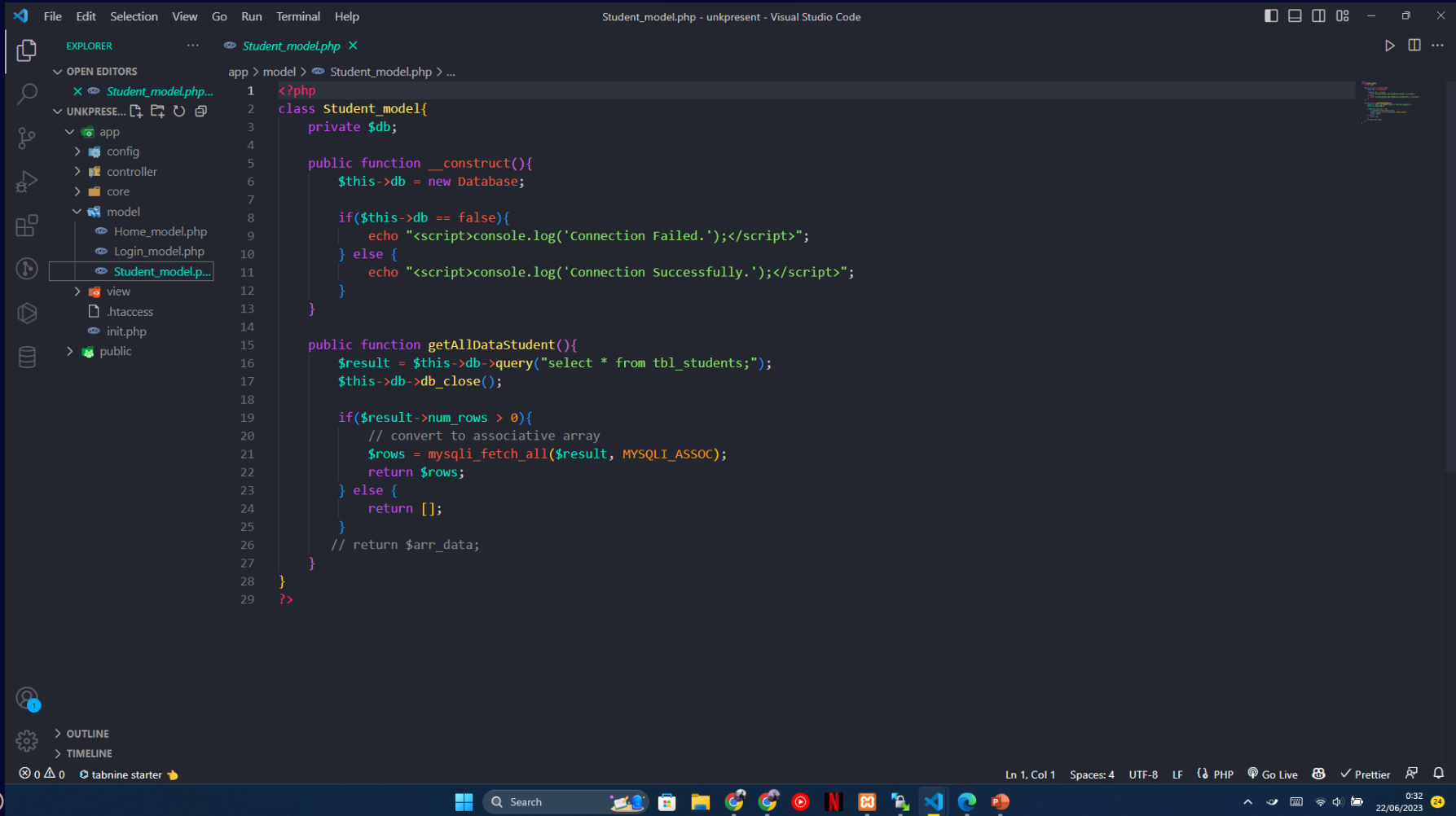
Buat file 'dosen.php' di dalam directory app/view/home



Buat file baru 'Student_model.php' di dalam directory app/model



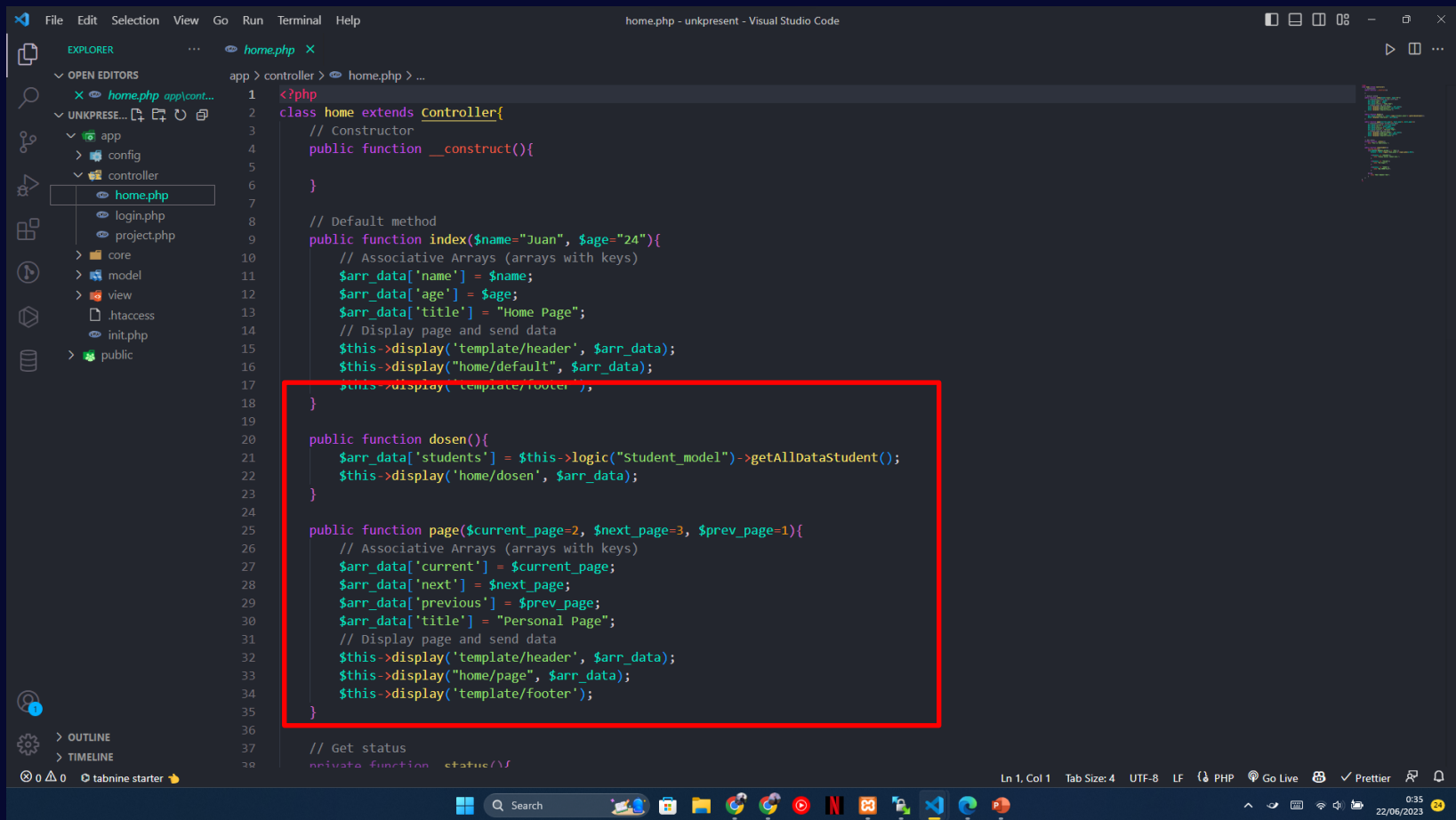
Tambahkan Code student_model.php



```
1 <?php
2 class Student_model{
3     private $db;
4
5     public function __construct(){
6         $this->db = new Database;
7
8         if($this->db == false){
9             echo "<script>console.log('Connection Failed.');
```

Student_model.php ini berguna untuk mengambil semua data dalam table 'tbl_students'

Add Function to Controller "Home.php"



```
<?php
class home extends Controller{
    // Constructor
    public function __construct(){

    }

    // Default method
    public function index($name="Juan", $age="24"){
        // Associative Arrays (arrays with keys)
        $arr_data['name'] = $name;
        $arr_data['age'] = $age;
        $arr_data['title'] = "Home Page";
        // Display page and send data
        $this->display('template/header', $arr_data);
        $this->display("home/default", $arr_data);
        $this->display('template/footer');
    }

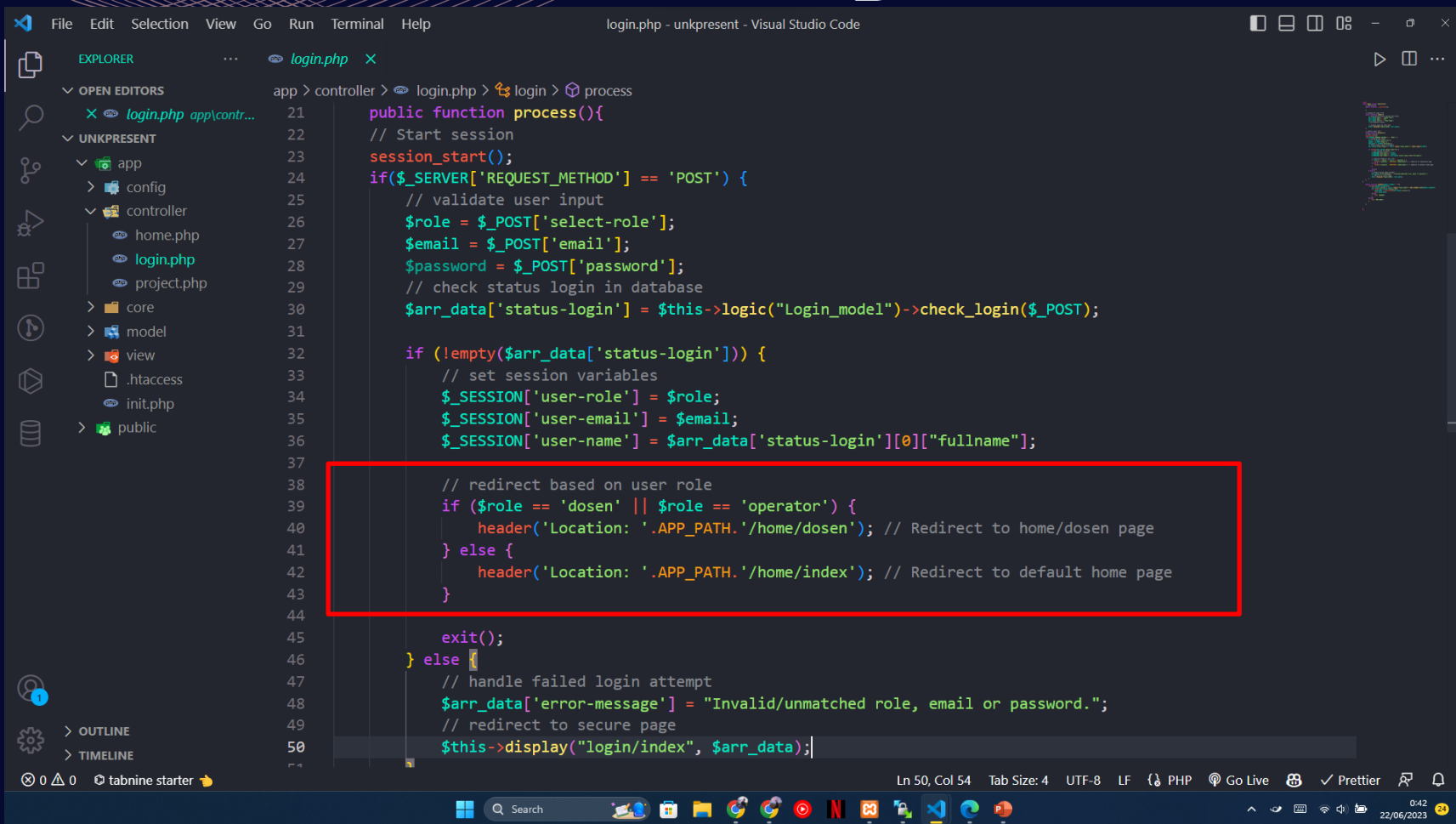
    public function dosen(){
        $arr_data['students'] = $this->logic("Student_model")->getAllDataStudent();
        $this->display('home/dosen', $arr_data);
    }

    public function page($current_page=2, $next_page=3, $prev_page=1){
        // Associative Arrays (arrays with keys)
        $arr_data['current'] = $current_page;
        $arr_data['next'] = $next_page;
        $arr_data['previous'] = $prev_page;
        $arr_data['title'] = "Personal Page";
        // Display page and send data
        $this->display('template/header', $arr_data);
        $this->display("home/page", $arr_data);
        $this->display('template/footer');
    }

    // Get status
    private function status($s
```

Function ini akan membuat kita pergi ke page dosen yang telah dibuat tadi (app/view/home/dosen.php) yang akan menampilkan semua data students

Tambahkan code di 'login.php'



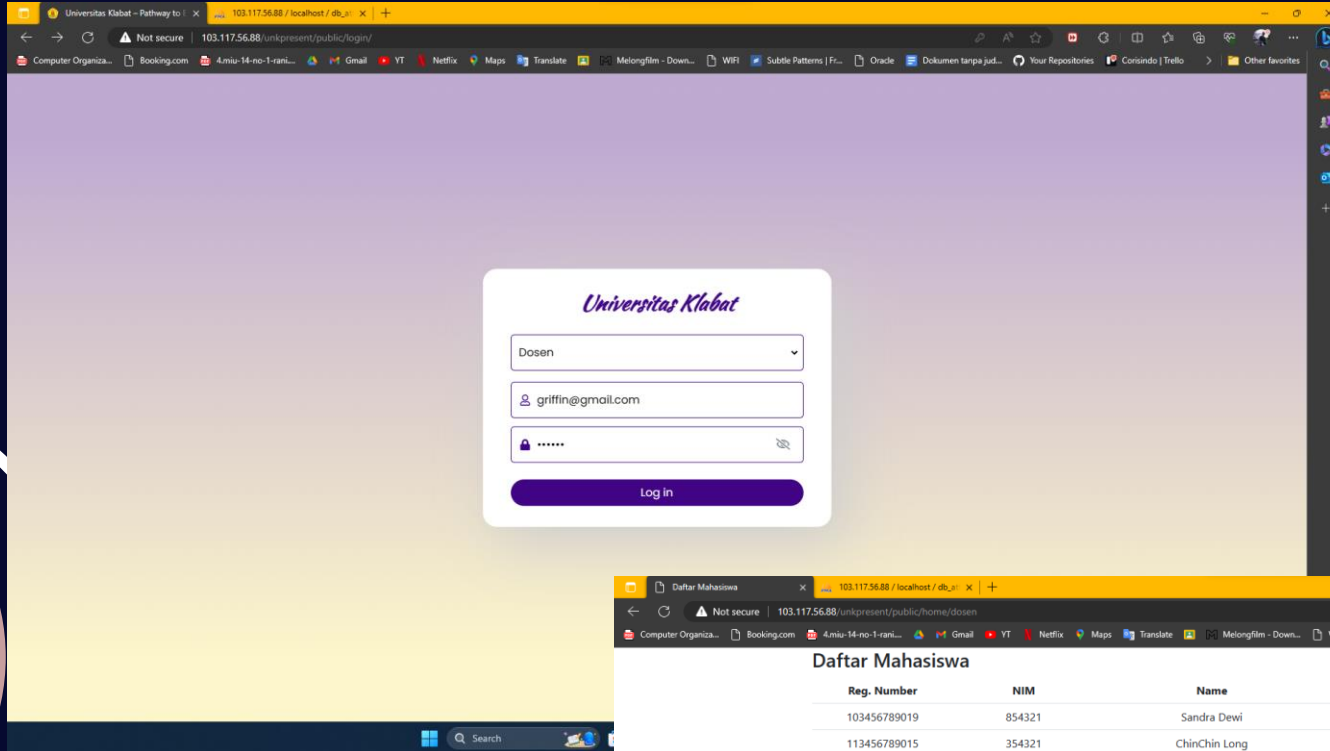
```
login.php - unkpresent - Visual Studio Code

app > controller > login.php > login > process

21 public function process(){
22     // Start session
23     session_start();
24     if($_SERVER['REQUEST_METHOD'] == 'POST') {
25         // validate user input
26         $role = $_POST['select-role'];
27         $email = $_POST['email'];
28         $password = $_POST['password'];
29         // check status login in database
30         $arr_data['status-login'] = $this->logic("Login_model")->check_login($_POST);
31
32         if (!empty($arr_data['status-login'])) {
33             // set session variables
34             $_SESSION['user-role'] = $role;
35             $_SESSION['user-email'] = $email;
36             $_SESSION['user-name'] = $arr_data['status-login'][0]['fullname'];
37
38             // redirect based on user role
39             if ($role == 'dosen' || $role == 'operator') {
40                 header('Location: '.APP_PATH.'/home/dosen'); // Redirect to home/dosen page
41             } else {
42                 header('Location: '.APP_PATH.'/home/index'); // Redirect to default home page
43             }
44
45             exit();
46         } else {
47             // handle failed login attempt
48             $arr_data['error-message'] = "Invalid/unmatched role, email or password.";
49             // redirect to secure page
50             $this->display("login/index", $arr_data);
51         }
52     }
53 }
```

Tambahkan code untuk memeriksa apakah user login menggunakan akun dosen atau operator . Jika iya, user akan langsung pergi ke halaman dosen.php yang mana akan ditampilkan data semua mahasiswa. Dan jika tidak, user akan ke halaman home yang umum

Output Ketika Login akun dosen



The screenshot shows a web browser window with the URL `103.117.56.88/unkpresent/public/home/dosen`. The page displays a table titled 'Daftar Mahasiswa' (Student List) with the following data:

Reg. Number	NIM	Name	Email	Password
103456789019	854321	Sandra Dewi	s854321@student.unklab.ac.id	12345
113456789015	354321	ChinChin Long	s354321@student.unklab.ac.id	12345
114456789015	644321	Sempoyonangan Ojomiso	s644321@student.unklab.ac.id	12345
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